Sequence 6

FUNCTIONS: REPRESENTATIONS AND REFERENCE FUNCTIONS

23 November–4 December 2026

I Function, image and preimage

Definition

A real-valued function associates to each real number x of its domain D_f a unique real number $f(x)$, called the image of x. A preimage of y is a real number $x \in D_f$ such that $f(x) = y$. We write $f : x \mapsto f(x)$. A number may have no preimage, one, or several.

Example 6.1

Let $g(x) = x^2 - 4x + 32$ on $\mathbb{R}$. Calculate $g(-5)$. Let $h(x) = 3x + 5$; determine the preimage of 6 under h.

Solution:

$g(-5) = 25 + 20 + 32 = 77$: 77 is the image of -5, and -5 is a preimage of 77. For h, $3x + 5 = 6 \iff x = 1/3$. The only preimage is $1/3$.

Example 6.2

For the square function, calculate the images of -3 and 3, then the preimages of 9, 0 and -1.

Solution:

$f(-3) = f(3) = 9$. The preimages of 9 are $-3, 3$; 0 has a single preimage, 0; -1 has none in $\mathbb{R}$.

```
def g(x):
    return x*x - 4*x + 32
```

Example 6.3

Distinguish `g(-5)` from `x = -5`.

Solution:

The first calls the function and returns 77. The second assigns -5 to the variable `x`, without calculating the image under g.

II Domain of definition

Conditions for existence

D_f contains the values for which the process defining f makes sense. A denominator must be nonzero; a number under a square root must be positive or zero. Also respect any domain imposed by the statement or the context.

Example 6.4

Determine the domains of definition:

$$f(x) = 3x^2 - 5x + 2, \quad g(x) = \frac{2x+1}{x-4}, \quad h(x) = \frac{x}{x^2-9}, \quad k(x) = \sqrt{2x-6}.$$

Solution:

$D_f = \mathbb{R}$. $D_g =]-\infty; 4[\cup]4; +\infty[$ since $x - 4 \neq 0$. $D_h =]-\infty; -3[\cup]-3; 3[\cup]3; +\infty[$ since $x^2 - 9 \neq 0$. $D_k = [3; +\infty[$ since $2x - 6 \geq 0$.

Watch out. The square root of a negative number does not exist in $\mathbb{R}$; it is not "a negative square root" that is excluded.

Example 6.5: Modelling

A rectangle of perimeter 20 m has width x m. Express its area and state the domain if both dimensions must be strictly positive.

Solution:

The length is $10 - x$, so $A(x) = x(10 - x)$ in m^2. $x > 0$ and $10 - x > 0$ give $D_A =]0; 10[$. If degenerate rectangles were allowed, we could take $[0; 10]$.

III Table of values and representative curve

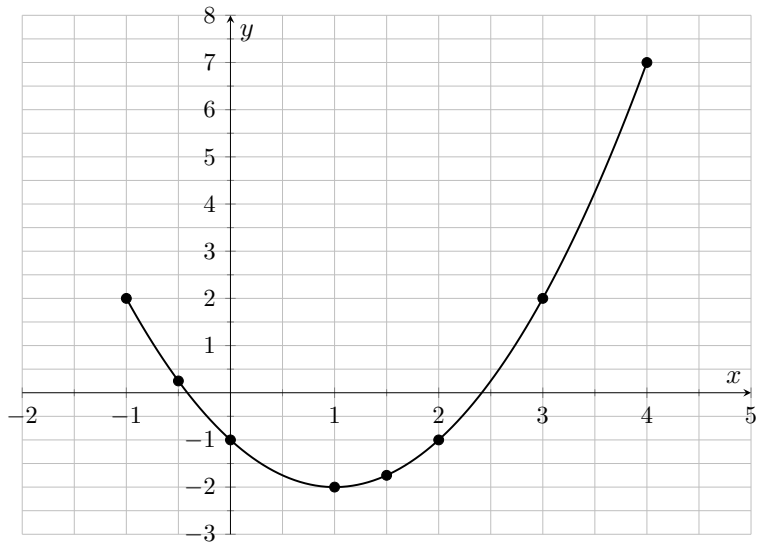
Definition

$\mathcal{C}_f$ is the set of points $(x; f(x))$, with $x \in D_f$. Thus $M(a;b) \in \mathcal{C}_f \iff a \in D_f$ and $b = f(a)$.

Example 6.6

On $[-1;4]$, $f(x) = x^2 - 2x - 1$. Complete the table, then draw the curve over the whole interval.

x	-1	-0.5	0	1	1.5	2	3	4
$f(x)$	2	0.25	-1	-2	-1.75	-1	2	7



Solution:

Images: 2; 0.25; -1; -2; -1.75; -1; 2; 7. Plot the points then draw the parabola. Segments joining only these points would give just an approximation.

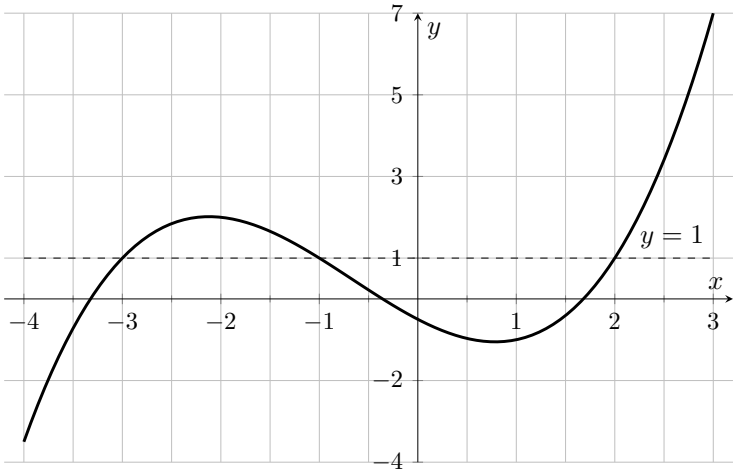
IV Graphical solving of equations and inequalities

Method

The solutions of $f(x) = k$ are the abscissae of the intersections of $\mathcal{C}_f$ with $y = k$. For $f(x) \leq k$, read the abscissae where the curve is below this line, intersections included; with $<$, the intersections are excluded.

Example 6.7

The curve represents $f(x) = 1 + \frac{1}{4}(x+3)(x+1)(x-2)$ on $[-4;3]$. Solve graphically $f(x) = 1$, $f(x) \leq 1$ and $f(x) > 1$.



Solution:

$f(x) = 1$: $-3, -1, 2$. $f(x) \leq 1$: $[-4;-3] \cup [-1;2]$. $f(x) > 1$: $]-3;-1[\cup]2;3]$. The endpoints $-4, 3$ come from the domain. The factorisation checks the intersections exactly; the algebraic study of the sign will be consolidated in sequence 8.

Teacher notes

Graph rebuilt to replace an ambiguous reading in the original booklet. Distinguish graphical conjecture from exact validation.

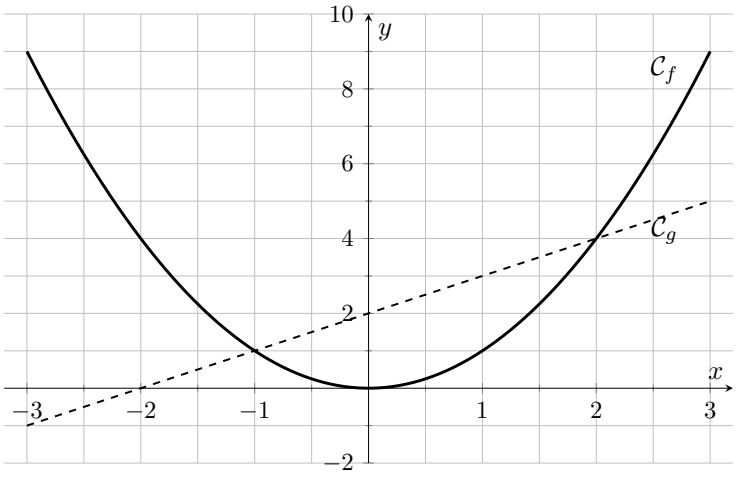
V Comparing two functions; reading a sign

Method

$f(x) = g(x)$ is read at the intersections of the curves. For $f(x) < g(x)$, $\mathcal{C}_f$ is strictly below $\mathcal{C}_g$ at the same abscissa. The sign of $f(x)$ is read relative to the x-axis.

Example 6.8

On $[-3;3]$, $f(x) = x^2$ and $g(x) = x + 2$. Solve $f = g$, $f < g$, $f \geq g$ and determine the sign of g.



Solution:

$f = g$ for $x = -1, 2$; $f < g$ on $]-1;2[$; $f \geq g$ on $[-3;-1] \cup [2;3]$. g is strictly negative on $[-3;-2[$, zero at -2, strictly positive on $]-2;3]$. Sign and variations are different pieces of information.

VI Affine functions and absolute value

Definitions

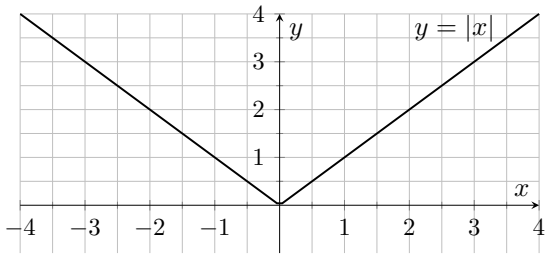
An affine function is $f(x) = mx + p$ on $\mathbb{R}$. It is linear if $p = 0$, constant if $m = 0$. For $m \neq 0$, it vanishes at $-p/m$; its sign is that of m to the right of this number. The function $x \mapsto |x|$ is defined on $\mathbb{R}$, positive or zero, and vanishes only at 0.

Example 6.9

Classify (several answers possible) as affine, linear, constant: $5x + 2$; $5x$; $7 + 2x - 7$; 6; $6(4x - 2)$; $6x + 5 - 6x$.

Solution:

All are affine. $5x$ and $2x$ are linear; 6 and 5 are constant; $24x - 12$ is neither linear nor constant.



Equations and inequalities

If $k < 0$, $|x| = k$ and $|x| \leq k$ have no solution, and $|x| \geq k$ is true on $\mathbb{R}$. If $k = 0$, the first two give $x = 0$, the third is true on $\mathbb{R}$. For $k > 0$:

$$|x| = k \iff x \in \{-k; k\}, \quad |x| \leq k \iff x \in [-k; k],$$

$$|x| \geq k \iff x \in]-\infty; -k] \cup [k; +\infty[.$$

Example 6.10

Solve $|x| = 5$, $|x| = -3$, $|x| \leq 3$, $|x| \geq 4$.

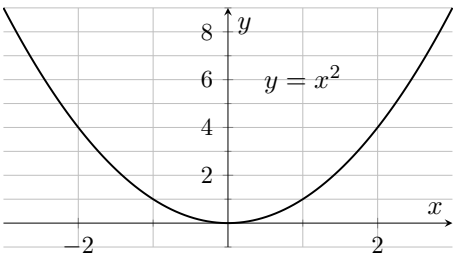
Solution:

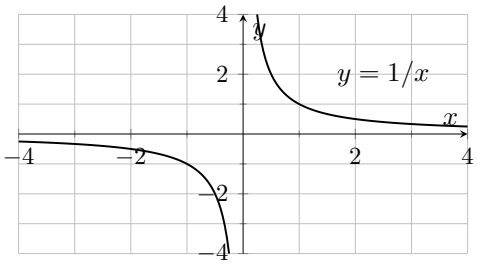
$\{-5; 5\}$; $\varnothing$; $[-3; 3]$; $]-\infty; -4] \cup [4; +\infty[$.

VII Square and inverse functions

Definitions and sign

$x \mapsto x^2$ is defined on $\mathbb{R}$; its curve is a parabola. $x^2 \geq 0$, with equality only at 0. $x \mapsto 1/x$ is defined on $\mathbb{R}^*$; its curve is a hyperbola. $1/x$ has the sign of x and never vanishes.





Equations and inequalities with x^2

If $k < 0$, $x^2 = k$ and $x^2 \leq k$ have no solution; $x^2 \geq k$ is true on $\mathbb{R}$. If $k = 0$, the first two give $x = 0$, the third is true on $\mathbb{R}$. For $k > 0$:

$$x^2 = k \iff x \in \{-\sqrt{k}; \sqrt{k}\}, \quad x^2 \leq k \iff x \in [-\sqrt{k}; \sqrt{k}],$$

$$x^2 \geq k \iff x \in]-\infty; -\sqrt{k}] \cup [\sqrt{k}; +\infty[.$$

Example 6.11

Solve $x^2 = 4$, $x^2 = -1$, $x^2 \geq 4$, $x^2 \geq -1$, $1/x = 4$, $1/x = 0$.

Solution:

$\{-2; 2\}$; $\varnothing$; $]-\infty; -2] \cup [2; +\infty[$; $\mathbb{R}$; $\{1/4\}$; $\varnothing$. More generally, $1/x = k$ has the unique solution $1/k$ if $k \neq 0$, none if $k = 0$.

VIII Inequalities with the inverse function

Method

Respect $x \neq 0$ and read both branches of the hyperbola. Do not multiply an inequality by x without knowing its sign.

	$k < 0$	$k = 0$	$k > 0$
$1/x \leq k$	$[1/k; 0[$	$]-\infty; 0[$	$]-\infty; 0[\cup[1/k; +\infty[$
$1/x \geq k$	$]-\infty; 1/k]\cup]0; +\infty[$	$]0; +\infty[$	$]0; 1/k]$

Example 6.12

Solve $1/x \geq 2$, $1/x \leq -1$, $1/x < 1/2$ and $1/x > -2$.

Solution:

$]0; 1/2]$; $[-1; 0[$; $]-\infty; 0[\cup]2; +\infty[$; $]-\infty; -1/2[\cup]0; +\infty[$. A strict inequality excludes the intersection with $y = k$; 0 is always forbidden.

Example 6.13

A student writes: "$1/x \leq 2 \iff 1 \leq 2x$, so $x \geq 1/2$". What is missing?

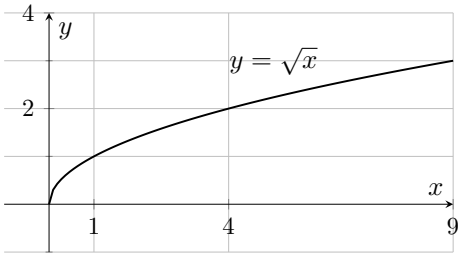
Solution:

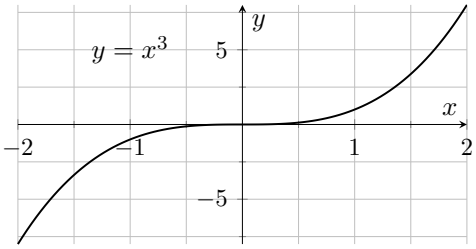
They implicitly assumed $x > 0$. For $x < 0$, $1/x < 0 < 2$: these values also work. The correct set is $]-\infty; 0[\cup[1/2; +\infty[$. The number -1 refutes their equivalence.

IX Using square root and cube

Definitions

$x \mapsto \sqrt{x}$ is defined on $[0;+\infty[$ and takes values that are positive or zero. For $k \geq 0$, $\sqrt{x} = k \iff x = k^2$; for $k < 0$, no solution. $x \mapsto x^3$ is defined on $\mathbb{R}$. Every real number k has a unique preimage, written $\sqrt[3]{k}$.





Example 6.14

Solve $\sqrt{x} = 3$; $\sqrt{x} = -2$; $\sqrt{x} \leq 2$; $\sqrt{x} > 3$; $x^3 = -8$; $x^3 \leq 27$.

Solution:

$\{9\}$; $\varnothing$; $[0;4]$; $]9;+\infty[$; $\{-2\}$; $]-\infty;3]$. Respect the domain for square roots. The cube function preserves order.

Example 6.15

Compare x^2 and x for $x = 1/2$, then $x = 2$. Is the square of a positive number always greater than the number itself?

Solution:

$1/4 < 1/2$ but $4 > 2$. The claim is false; $1/2$ is a counterexample. General study in sequence 10.